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P2 -Plotting oceanographic data and timeseries

Part 1- profiles

1.

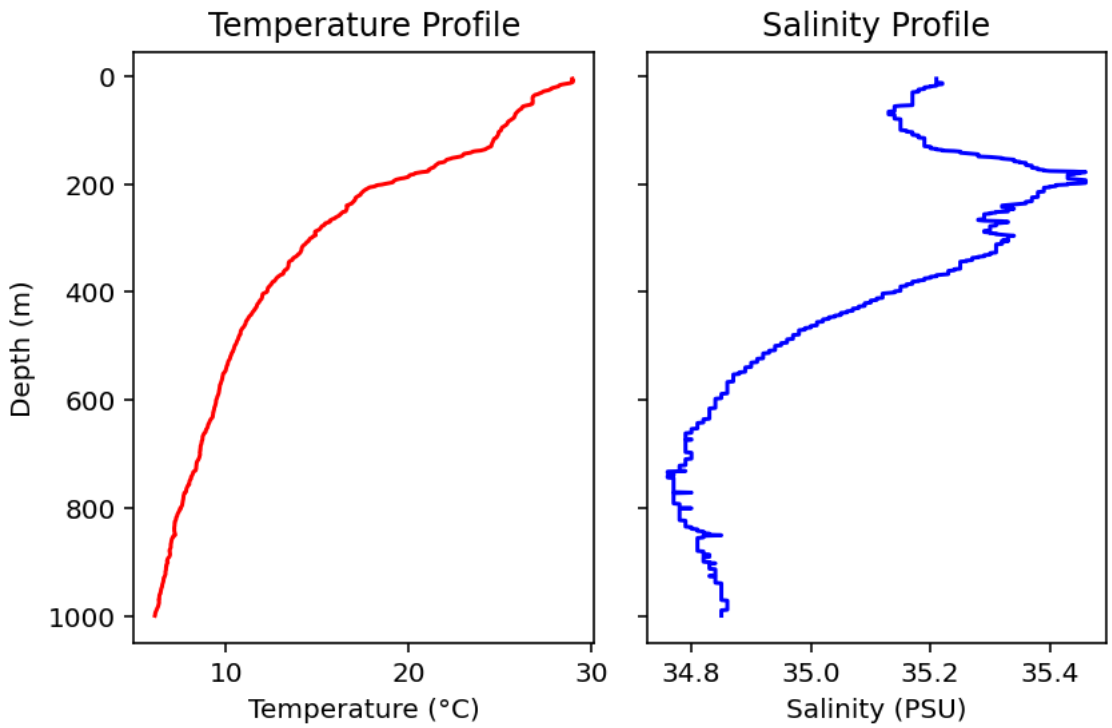

Figure 1: Temperature and salinity profiles plotted against depth from the CTD dataset. The data set is named as for_nitpicker.dat from a CTD, the primary instrument used by oceanographers. The left panel shows the temperature profile (red line) and the right panel shows the salinity profile (blue line), allowing comparison of the vertical variation of temperature and salinity in the water column to a depth of 1000m.

PART 2- Time series plots, Histogram, statistics and scatter plot

1. Time series plot

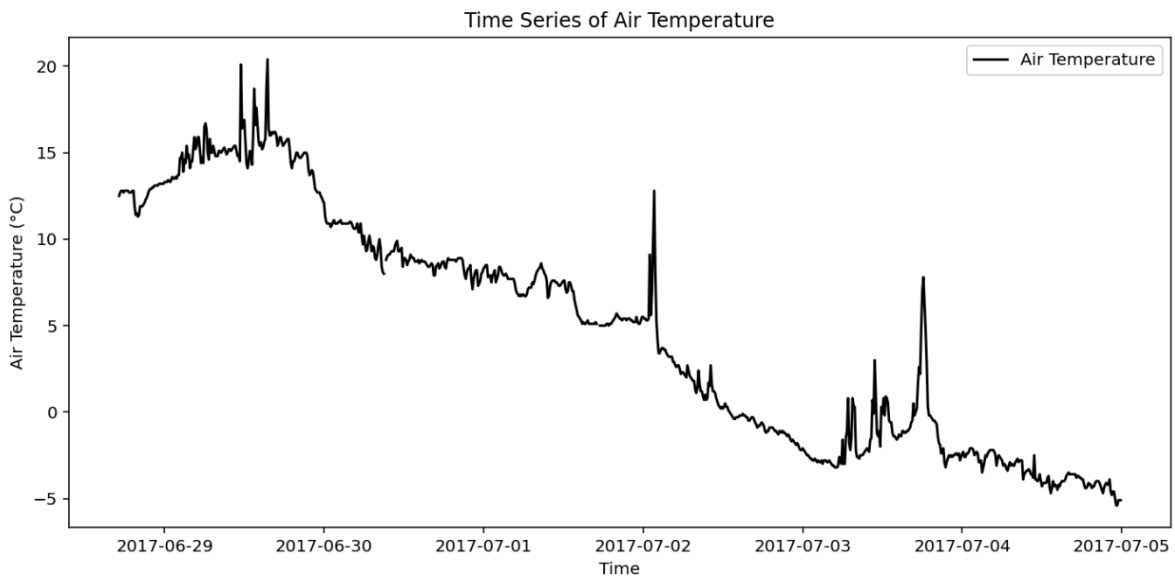

Figure 2: Time series of air temperature recorded from the metocean dataset at 10-minute intervals, showing temperature variations from 29 June 20217 to 4 July 2017. The plot highlights a gradual cooling trend over the period, with several sharp spikes likely corresponding to weather events that might have occur during this period.

2. Histogram graph

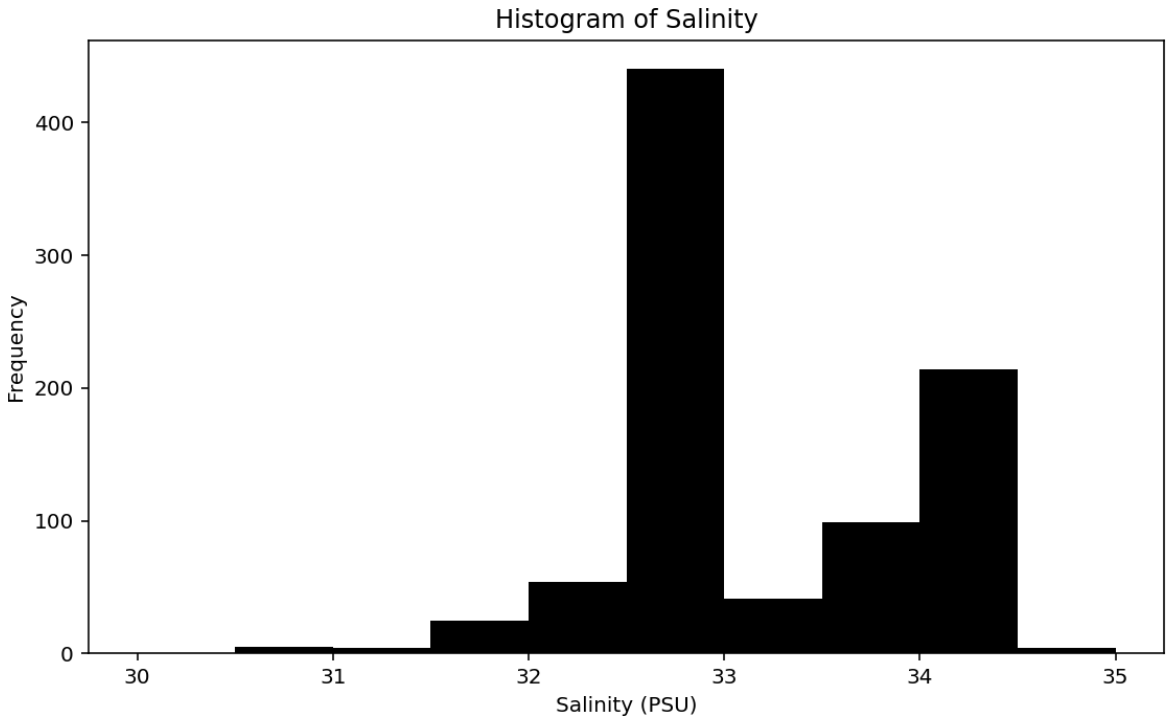

Figure 3: The histogram illustrates the variability and overall distribution of salinity. Seawater salinity (PSU) displays the frequency distribution of salinity values. The x-axis represents salinity (PSU), while the y-axis shows the frequency of observations using bins of 0.5 PSU between 30 and 35. A high frequency of salinity was observed between approximately 32.5 and 34.5 PSU, indicating that majority of the sampled salinity values are within this range.

3. Summary statistics

Table 1: Summary statistics for air temperature (°C) and salinity (PSU) from the selected dataset up to 4 July 2017. The table shows the mean, standard deviation, and interquartile range (IQR).

Variables	Mean	Standard deviation	IQR
Temperature	4.878271	6.692847	11.675000
Salinity	33.110553	1.140150	1.272575

4. Scatter plots

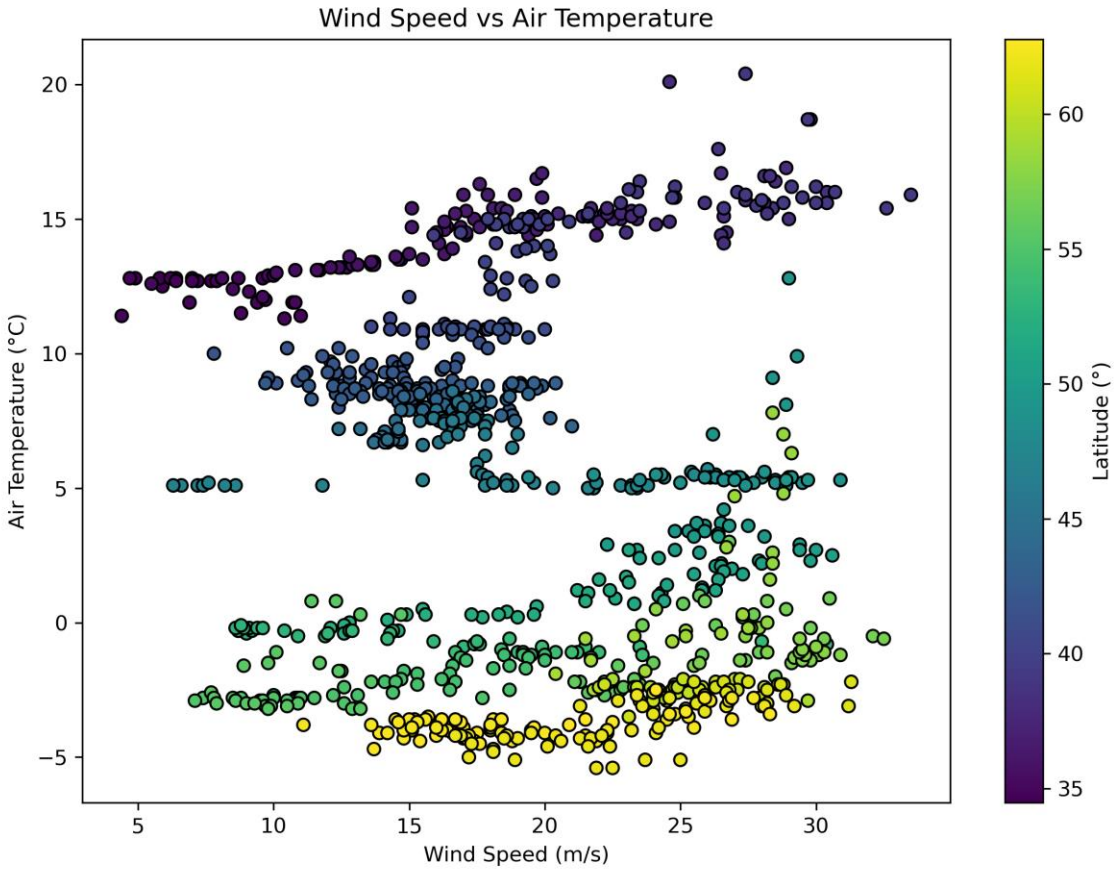

Figure 4: Scatter plot showing the relationship between wind speed (m/s) and air temperature (°C). The x-axis represents wind speed(mls), while the y-axis shows air temperature(°C). Each point represents an observation, and the colour scale indicates latitude (°). The plot illustrates how air temperature varies with wind speed across different latitudes, with higher latitudes associated with lower air temperatures.